Project goals: Small, repeatedly-flooded municipalities with older building stock have no engineering tool to evaluate whether community-scale drainage investment or distributed building hardening better reduces flood damage. Here we propose a closed-loop cyber-physical system (CPS) that couples the physical behavior of pre-code load-bearing masonry and timber, absent from existing fragility frameworks, with a computational core that turns free national data and community memory into building-level damage estimates. It senses through USGS gauges and NOAA precipitation operationally, with commercial ICEYE Synthetic Aperture Radar (SAR) for one-time building-level duration and depth calibration and open Sentinel-1 SAR for flood-extent context; its computational core is a Bayesian uncertainty-quantification framework fusing a GIS-parameterizable mass balance model with community-elicited duration priors and a gauge-stage-to-terrain projection for building-level depth; and the community closes the loop through a co-developed three-mode interface for retrofit planning, pre-storm staging, and post-event updating. **RQ1** (Napolitano, Wauthier, and Busse) asks to what extent this CPS, operating on free national data and community knowledge, can produce building-level damage assessments decision-adequate to weigh community-scale drainage against building-scale hardening on a common probabilistic basis, without hydraulic simulation. **RQ2** (Napolitano with input from all PIs) asks under what conditions locally held community knowledge, entered as Bayesian calibration priors, improves damage assessments relative to a GIS-only baseline, and where it functions as a substitute versus complement to geospatial data. The intellectual contribution is the closed loop these questions form, in which free national data and community knowledge together support real investment decisions, with community knowledge a structural calibration input measured against a geospatial baseline.

1 Status quo: Small communities at the confluence of riverine flood hazard and older building stock face a resilience challenge that no existing engineering framework is designed to address. Across the northeastern United States and Appalachian corridor, thousands of towns share this profile of multi-day riverine flood exposure, older load-bearing masonry and timber cores, and institutional capacity below the threshold for hydrodynamic simulation [1, 2]. Of the more than 22,000 communities FEMA's National Flood Insurance Program enrolls nationwide [3], thousands in this region combine significant pre-1940 building stock, no in-house hydraulic modeling capacity, and repeated riverine flooding. In Vermont alone, over a quarter of the housing stock predates 1939 [4, 5]. For these communities, every flood event surfaces the same unresolved question. Where do limited flood-mitigation dollars belong?

These communities face a menu of mitigation choices, most of them permanent investments and a few made in the hours before a storm. What they lack is any way to weigh those choices on a common probabilistic footing, using the assets they actually have, a handful of physical sensors, some public geospatial data, and a deep store of community memory. The sharpest case is the allocation question. Would a community-scale drainage investment, such as a retention basin, reduce damage enough to justify its cost over the same dollars spent hardening individual buildings? Answering it means linking an upstream drainage decision to downstream building damage probabilities, a comparison no existing framework lets a small community make.

That comparison demands two things these communities do not have. The first is a model. Hydrodynamic simulation pipelines such as HEC-RAS and LISFLOOD-FP require calibrated hydraulic models, high-resolution topographic data, and sustained technical staff that small towns cannot maintain [6–8]. The second is data on how their own buildings fail. Damage functions in HAZUS and FEMA P-58 are built around post-1945 construction, so the load-bearing masonry and timber structures built before modern building codes (pre-code), the dominant type in repeatedly flooded historic downtowns, are absent from existing fragility frameworks [9, 10]. Those frameworks also treat damage as a function of depth alone, not the extended-inundation degradation that governs older masonry, which we review in the fragility state of the art below. Even the most authoritative federal guidance on functional recovery, NIST SP 1310 [11], concedes that its framework is too complex for smaller agencies with limited resources and funding, and calls for simplifications that smaller organizations can actually implement [11, Sec. 6.1.3]. The result is a dead end. Emergency managers, preservation officials, and building owners have no engineering tool that turns a flood event into building-specific guidance.

That dead end has one exception. What these towns lack in models and data, they hold in memory. Residents know how deep the water rose, how long it stayed, which buildings took the worst of it, and how often it returns. Existing community resilience frameworks are largely top-down, built around standardized data collection and modeling pipelines that resource-constrained localities cannot sustain, and none gives a community a way to put this knowledge into an engineering-grade probabilistic assessment. We treat that knowledge not as context but as a calibration input, locally held flood memory entered as Bayesian priors, whose contribution to a damage assessment can be measured directly against a baseline built from geospatial data alone. Building that measurement into a working loop, from free national data and community memory to a damage estimate a town can act on, is the work that follows.

1.1 Research questions and intellectual merit:

- **RQ1 (Napolitano, Wauthier, and Busse):** To what extent can a CPS operating on free national data and community knowledge, without hydraulic simulation, produce building-level damage assessments decision-adequate to weigh community-scale drainage against building-scale hardening on a common probabilistic basis?
- **RQ2 (Napolitano with input from all PIs):** Under what conditions does locally held community knowledge, entered as Bayesian calibration priors, improve flood damage assessment accuracy and calibration relative to a GIS-only baseline, and where does it function as a substitute versus complement to geospatial data?

The central contribution is the closed loop these questions form, a working cyber-physical system in which free national data and locally held community memory together produce building-level damage assessments decision-adequate for real investment choices, with no hydraulic simulation anywhere in the loop. Realizing it required three advances that did not exist for this setting. The first is **computational**: a Bayesian framework that infers building-level inundation depth and duration from sparse, indirect, interval-censored observations (gauge stage projected across terrain, SAR duration brackets), fuses them with a GIS-parameterizable mass balance model and community-elicited priors, and propagates uncertainty to a decision-adequate damage estimate. The second is **physical**: the CPS couples to its environment on both ends, sensing physical flood signatures (inundation depth, how long the water stayed, resulting damage) and informing physical interventions (drainage retrofits that shorten a sub-basin's drainage timescale, building hardening that raises the masonry capacity threshold) whose effects reenter future flood behavior. The third is treating locally held **community knowledge** as a structural calibration input: a characterization of when knowledge entered as Bayesian priors improves accuracy and calibration over a GIS-only baseline, and where it functions as a substitute versus a complement to geospatial data. Together these establish a generalizable principle, that free national data fused with community knowledge is sufficient for the investment decisions small municipalities face, transferable to any domain with national data and community knowledge but no simulation capacity.

2 Current state-of-the-art:

Four literature domains each supply one piece of the loop this proposal closes: a hydrological prior on how long water stays, a physical damage model for the affected building stock, a mechanism for entering community knowledge as quantified belief, and a decision framework for allocating mitigation dollars. None couples them, so the loop has never been evaluated end to end. No prior work runs free national data and community knowledge through a physical model of pre-code building stock to a damage estimate a municipality acts on, then feeds those actions and each new event back into the model. That is what RQ1 and RQ2 test, arbitrated by real flood outcomes across 271 buildings with sensor-measured durations and field-recorded damage states.

The broader CPS literature does not close it either. Community sensing architectures for flood monitoring assume local sensor infrastructure, proprietary data contracts, or sustained community effort that resource-constrained municipalities cannot maintain after a grant ends, and none carries the community knowledge mechanism, independent ground truth, or physical damage model the loop requires. This proposal's free national inputs, USGS stream gauges, NOAA precipitation, and SAR imagery, make it tractable and transferable to any gauged municipality, with community knowledge entering the computational core as measurable Bayesian priors rather than supplementary narrative.

2.1 Hydrological mass balance modeling and flood duration estimation: Hydrological mass balance models estimate flood volume, storage, and drainage timescale from watershed geometry, land cover, soil properties, and precipitation inputs. Unit hydrograph methods and rational method variants provide event-scale peak discharge and runoff volume estimates for small watersheds from tabulated curve numbers and contributing area [12]; GIS-parameterized implementations using USGS StreamStats, NHD Plus, NLCD, and SSURGO extend these approaches to ungauged and lightly gauged basins without local calibration data [13, 14]. Prior work has applied these methods to estimate peak discharge and flood extent at community scale; their application to building-level inundation duration estimation, the input required by duration-dependent fragility models, has not been characterized. Whether a GIS-only mass balance model can produce building-scale drainage timescale estimates accurate enough for retrofit allocation where calibrated hydraulic models are unavailable is unknown (Subtask 1.2).

2.2 Fragility and vulnerability frameworks for flood damage: Flood fragility and vulnerability assessment relies predominantly on depth-damage functions that map peak inundation depth to expected repair cost or damage ratio by occupancy class. The HAZUS flood technical manual provides the most widely used depth-damage curves for the United States, organized by building occupancy and foundation type [9]; comparable databases have been developed in Europe and Australia for loss estimation in community-scale flood events [15, 16]. These functions share a structural assumption. Peak depth is the governing mechanical input, and damage accumulates instantaneously at flood peak rather than over the inundation period. For post-1945 wood-frame and light steel construction that assumption is approximately correct; for pre-code load-bearing masonry and timber construction it is not. Extended inundation drives duration-dependent degradation, including mortar softening, progressive brick saturation, and rising damp, that depth-damage functions cannot capture [17–19]. Observational studies of masonry buildings in European riverine flood events have documented that damage state is better predicted by depth and duration together than by depth alone [20–22], yet no duration-dependent fragility surface exists for the pre-code load-bearing masonry archetypes that dominate repeatedly flooded northeastern American historic cores. This absence blocks CPS integration. A duration-producing computational core needs duration-dependent fragility, which no existing framework provides for this building stock.

2.3 Community knowledge in probabilistic risk assessment: Community knowledge, local observation, and resident experience constitute a substantial information resource for disaster risk assessment that formal modeling pipelines largely bypass. The Sendai Framework for Disaster Risk Reduction explicitly prioritizes community-based approaches [23]; community-based disaster risk reduction programs have demonstrated that locally held knowledge can identify hazard exposures, vulnerable populations, and infrastructure conditions that national datasets do not capture [24]. In flood risk specifically, participatory GIS methods and volunteered geographic information platforms have been used to map inundation extents, document damage, and supplement remotely sensed observations [25, 26]. Existing approaches use community knowledge for post-hoc validation, qualitative prioritization, or supplementary narrative input; none operationalizes it as a structural, quantitative calibration input to a probabilistic building damage model. The Bayesian prior elicitation literature provides a principled mechanism for translating expert and stakeholder judgment into probability distributions [27, 28], shown elsewhere to improve model calibration when data are sparse, but not yet applied to community flood knowledge. Whether community knowledge and GIS data function as substitutes, complements, or redundant sources across building and drainage contexts remains entirely untested.

2.4 Flood decision support for resource-constrained municipalities: Federal frameworks address adjacent pieces of this problem in isolation: the FEMA benefit-cost toolkit evaluates individual mitigation projects against grant criteria [29]; the NIST Community Resilience Planning Guide supports capital planning [30, 31]; federal hazard mitigation grant programs govern drainage and structural investments without linking them to building damage probabilities [32]; and FEMA's flood retrofit guidance addresses building-by-building measures without a community-scale framework [33, 34]. None couples drainage investment outcomes to building damage probabilities through a physical model of the affected building stock, so nothing carries a completed retrofit back into the flood behavior a town should expect next time, and none allows community-scale drainage investment and distributed building hardening to be compared on the same probabilistic basis. None is designed for small municipalities with limited institutional capacity and no sustained technical staff; NIST SP 1310 explicitly acknowledges this limitation and calls for simplified analogs appropriate to smaller organizations [11, Sec. 6.1.3], and the hazard mitigation planning literature documents that plan quality and implementation capacity fall systematically with municipal size and rural status [1, 2, 35]. The fragility gap above propagates directly into these decision frameworks. None addresses pre-code masonry or duration-dependent damage states. No existing framework provides building-level inundation risk predictions from forecast precipitation for pre-storm emergency resource staging; flood decision support has been designed for post-event or inter-event planning contexts, not for the hours before a flood peak when pre-positioning decisions must be made. Nor does post-event practice feed back: assessments are reissued from static inputs rather than updated by what the last flood revealed, so no framework improves as a community accumulates events.

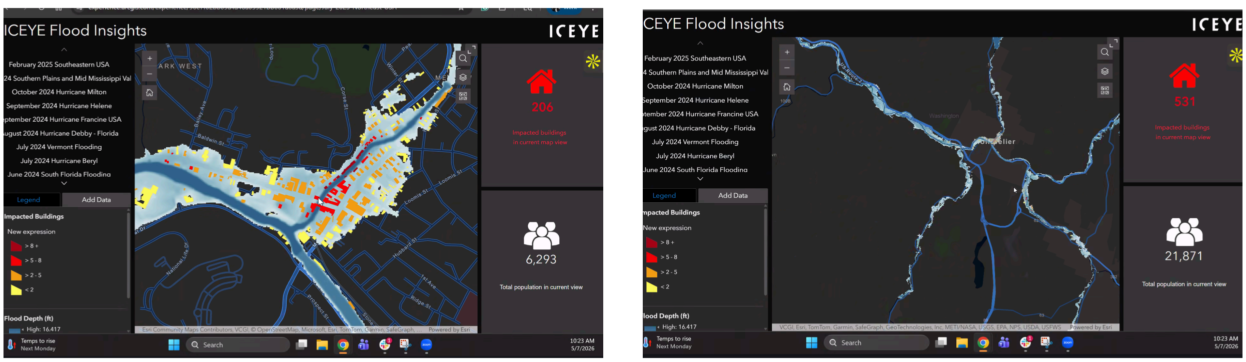
2.5 Gap summary: The pieces exist; the loop does not. Each domain above supplies one component and stops at the interface with the next: mass balance models are uncharacterized at building-level duration, fragility frameworks cover neither pre-code masonry nor duration-dependent damage, community knowledge has never been operationalized as a quantitative calibration input, let alone measured against a geospatial baseline, and no decision framework connects drainage investment to building damage probabilities. NIST SP 1310 [11] explicitly calls for the development of analogous frameworks for flood and severe storms as necessary future work [11, Sec. 6.1.2]; no existing tool answers that call for the building stock and community decision context this proposal targets. The research plan that follows closes the loop those components leave open.

3 Partner site and preliminary results: The CPS is developed, calibrated, and validated in Montpelier, Vermont across two flood events (2023 and 2024); PI Napolitano has been working directly with the Montpelier community since 2023. Montpelier, Vermont's capital, sustained catastrophic flooding in July 2023 (80% of its National Register Historic District damaged) and again in 2024, one of over 45 such events since 1900 at the Winooski/North Branch confluence. Its downtown core of 19th-century load-bearing masonry, largely pre-dating the 1933 ANSI code, is this proposal's pre-code archetype, with moderate institutional capacity (emergency manager, SHPO support, planning department) but no hydrodynamic simulation infrastructure. The framework transfers to any municipality meeting four conditions: riverine flooding with building-level standing water persisting roughly five or more days; pre-1940 construction as a meaningful share of the

Figure 1: Montpelier during the July 2023 flood.

flood-exposed inventory; no sustained in-house hydraulic modeling capacity; and USGS gauge coverage on the primary watershed. The first condition is defined on standing-water persistence in the building stock, not on the shorter river-stage exceedance (Preliminary Results below). It is this slow local drainage, not the river crest, that drives duration-governed masonry degradation and that the mass balance model targets.

3.1 Preliminary results: SAR building-level data, Montpelier 2023 and 2024: SAR imagery captures flood extent through cloud cover and at night, when optical imagery fails. Open Sentinel-1 (C-band, 10 m resolution) and commercial ICEYE (X-band, <1 m resolution) both cover the two Vermont events at the meter-scale, few-hour cadence this framework requires (Subtask 1.1). In particular, ICEYE commercial SAR provides building-level flood depth and inundation maps fusing SAR amplitude imagery with LIDAR and water gauge measurements, at revisit cadences as fine as six hours during dense acquisition sequences, across the inundation and recovery period of both Montpelier flood events. Through an NDA data access agreement, ICEYE data for 271 downtown pre-code masonry buildings, each with ICEYE-measured duration, peak depth, and field-recorded damage state, will be analyzed for both the 2023 and 2024 events (Subtask 1.1). The 271-building dataset constitutes the empirical foundation for fragility surface calibration (2023) and held-out validation (2024); the 2024 data is not used in any model fitting (Fig. 2).



(a) July 2023 (calibration event)

(b) July 2024 (held-out validation event)

Figure 2: ICEYE building-level flood depth, Montpelier downtown core, for both study events. Building footprints are colored by peak inundation depth (yellow: <2 ft; orange: 2–5 ft; red: 5–8 ft; dark red: >8 ft). Duration brackets, as fine as six hours between successive overpasses, were extracted separately for all 271 buildings across both events (Subtask 1.1).

3.2 Preliminary results: field damage assessment, Montpelier 2023:

3.3 Preliminary results: Winooski watershed drainage timescale: To test whether the mass balance timescale is recoverable from free gauge data at fragility-model accuracy, we ran a recession analysis of USGS gauge 04286000 (Winooski River at Montpelier; 397 mi^2) for both events using 15-minute discharge records (Fig. 3): fitting $Q(t) = Q_p\, e^{-t/\tau}$ to the first 60 hours of the falling limb gives $\tau = 37$ h ($R^2 = 0.91$) for July 2023 (peak 23,100 cfs, stage 21.3 ft, 6+ ft above the 15 ft NWS minor flood stage) and $\tau = 45$ h ($R^2 = 0.96$) for July 2024 (peak 11,900 cfs). Three observations anchor the research plan. First, the watershed-scale timescale (37, 45 h) is stable across events, at the day-to-multi-day order masonry degradation operates on, and recoverable from the gauge record alone; Subtask 1.2 disaggregates this quantity to sub-basin scale from national GIS layers. Second, the 2023 recession departs from a single exponential beyond roughly 60 hours (R^2 falls to 0.73), consistent with a channel-to-storage transition: standing water in low-lying sub-basins outlasts the river recession, motivating sub-basin $\hat{\tau}_k$ estimates over one watershed constant. Third, the events differ nearly twofold in peak magnitude, and the 2024 peak stage (14.5 ft) stayed below minor flood stage even as downtown parcels took water, so the 2023-calibrated, 2024-validated design tests transfer across event magnitudes rather than replicating one event size. Extending the analysis across the gauge's full 1990–2026 record strengthens this anchor. Fifteen well-fit events give $\hat{\tau}$ between 36 and 96 hours (median 51), $\mathrm{sd}(\ln \hat{\tau}) = 0.30$ (the event-to-event anchor σ_{event} for the Subtask 1.3 calibration), and a quantified antecedent-moisture signal, a roughly 33-hour shift from driest to wettest antecedent state, observable from the same gauge the operational system already reads. Discharge stayed above twice pre-event baseflow for roughly 19 days in 2023 and 10 in 2024, bounding the window within which building-level standing water drained, consistent with the framework's multi-day scope conditions. A stage-threshold analysis makes the standing-water distinction concrete. The 2023 gauge stood above NWS minor flood stage for 28 hours and above action stage for 47 hours, yet building interiors were still being pumped days later, the State Street office complex took four to five days to clear [36], the Montpelier High School basement five to six days [37], and downtown streets roughly two days [38]. Building-level standing water, the quantity T_i the framework models, thus outlasted river-stage exceedance by days in 2023, the separation sub-basin timescales $\hat{\tau}_k$ are designed to capture. A first prototype of the Subtask 1.2 stage-to-DEM projection (1 m 3DEP terrain; gauge datum 499.87 ft NAVD88) tests the depth pathway. A flat-surface projection of the 2023 crest reproduces river-corridor inundation ($\sim$92 hydraulically connected acres) but underpredicts the flooded core, with Main Street landmarks 0.8 to 1.5 m above the projected surface, 1.2 to 1.5 km upstream, implying a water-surface slope of 0.6 to 1.0 m/km, the single parameter the 2023 ICEYE depth field calibrates. A complementary sweep of that slope against documented 2023 outcomes (Main Street flooded, the State House dry) brackets it at 0.9 m/km or above, predicting 234 inundated buildings at that lower bound, the same order as the 271-building empirical set the ICEYE depth field pins precisely. The analysis scripts and gauge records are archived with the project repository for reproducibility; discharge values are USGS provisional.

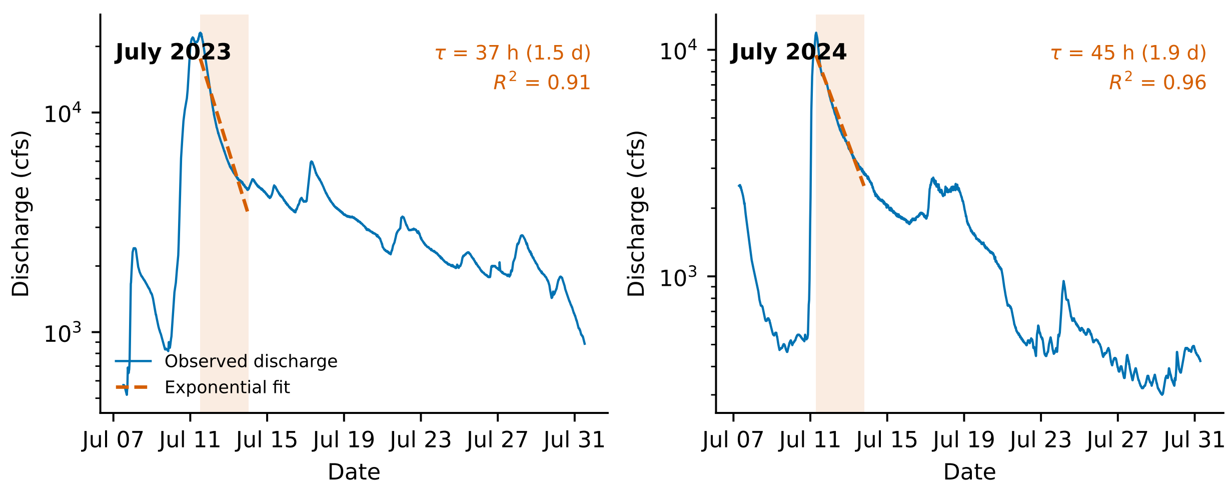


Figure 3: Recession analysis of USGS gauge 04286000 (Winooski River at Montpelier), July 2023 (left) and July 2024 (right), from 15-minute discharge records. Dashed lines show exponential fits over the shaded 60-hour windows: $\tau = 37\,\mathrm{h}$ ($R^2 = 0.91$) and $\tau = 45\,\mathrm{h}$ ($R^2 = 0.96$).

4 Research plan: RO1 (Napolitano, Wauthier, Busse) addresses RQ1 through five coupled subtasks (1.1 to 1.5); RO2 (Napolitano with all PIs) addresses RQ2 through four (2.1 to 2.4). Leads and dependencies are given in each subtask header below and in the Project Timeline section.

This CPS's dynamics, estimation, and control close at three timescales rather than through continuous automated actuation: hours before a flood peak (pre-storm staging from NOAA forecasts), per event (post-event updating of community priors and the error model), and years (retrofit actuation through the $\hat{\tau}_k$ and fragility updates of Subtask 2.4). Human decisions are the actuators in all three loops, so the community is a functional component of the control loop rather than a recipient of its outputs, the connected-communities reading of CPS the CIR track invites. The research questions emerged from PI Napolitano's post-2023 Montpelier engagement, whose partners' question, where limited flood-mitigation dollars belong, defined the decision problem.

Research Objective 1 (RO1, leads: Napolitano, Wauthier, and Busse): Develop and validate a closed-loop CPS coupling a GIS-parameterizable mass balance model with community knowledge in a Bayesian framework and duration-dependent fragility surfaces for pre-code masonry archetypes, and establish whether the system is decision-adequate for retrofit allocation and pre-storm resource staging without hydraulic simulation, calibrated on Montpelier 2023 and validated on the held-out 2024 event.

RO1 couples four components across Subtasks 1.1 to 1.4: building-level ICEYE SAR processing (with open Sentinel-1 for flood-extent context), a GIS-parameterizable mass balance model, a Bayesian fusion framework, and duration-dependent fragility surfaces. ICEYE provides the one-time calibration data, delivered as interval-censored duration brackets with characterized classification error, for both the framework's error model and the fragility surface; the operational system runs entirely on USGS gauge data and NOAA precipitation, which are free and nationally available. Subtask 1.5 validates the full RO1 system against the 2024 held-out event and characterizes decision-adequacy across all three operational modes.

Subtask 1.1: SAR Data Processing, Interpretation, and Likelihood Characterization (lead: Wauthier). Subtasks 1.1 and 1.2 run in parallel in Year 1 and are independent until both feed the Bayesian fusion framework in Subtask 1.3. PI Wauthier processes ICEYE commercial SAR data for the 271 downtown Montpelier buildings across both flood events, extracting building-level peak depth and per-overpass wet/dry status, and validating building footprint attribution against ground-referenced building inventory. Because duration is inferred from wet/dry status across successive overpasses, each building's inundation duration is delivered not as a point value but as a bracket between the last overpass observed wet and the first observed dry, with bracket width set by the acquired revisit cadence (as fine as six hours where the acquisition sequence is dense); this bracket is carried through the framework as an explicit measurement interval rather than collapsed to a false point estimate. The key scientific outputs of this subtask are the per-building duration brackets and peak depths for both events, together with per-scene wet/dry classification error rates; once Subtask 1.2 delivers $\hat{\tau}_k$ estimates, comparing mass balance predictions against the 2023 brackets calibrates the building-scale error variance σ^2_{MB} of the GIS-parameterized model in Subtask 1.3. An archived cadence simulation shows that scenes spread across the multi-day drainage window carry substantially more calibration information than scenes clustered near the peak, and this criterion prioritizes which acquisitions are processed. Open Sentinel-1 imagery is processed for both events for broader flood-extent context; at 10 m resolution it cannot resolve individual downtown buildings, so it supports cross-checking of the ICEYE classifications rather than feeding the building-level likelihood.

Subtask 1.2: GIS-Parameterizable Mass Balance Model (lead: Busse). PI Busse develops a hydrological mass balance model for the Winooski watershed above Montpelier, estimating the drainage timescale $\hat{\tau}_k$ for each drainage sub-basin from nationally available GIS layers: watershed delineation and contributing area from USGS 3DEP and NHD StreamStats, effective impervious fraction from NLCD, soil infiltration rates from SSURGO, and real-time discharge from USGS station 04286000. Each sub-basin is modeled as a linear reservoir, the storage-discharge form whose signature is the exponential recession observed in the preliminary gauge analysis:

$$\frac{dS_k}{dt} \;=\; I_k(t) \;-\; Q_k, \qquad Q_k \;=\; C_k\,\frac{S_k}{S_k^{\max}} \;=\; \frac{S_k}{\hat{\tau}_k}, \qquad \hat{\tau}_k \;=\; \frac{S_k^{\max}}{C_k} \tag{1}$$

where S_k is stored floodwater volume in sub-basin k, $I_k(t)$ is inflow from precipitation and upstream routing, $S_k^{\max}$ is the storage capacity of the sub-basin's low-lying terrain from 3DEP, and C_k is its effective conveyance and infiltration capacity from NHD channel properties, NLCD imperviousness, and SSURGO infiltration rates. Outflow is closed as a linear function of fractional storage, $Q_k = C_k\,S_k/S_k^{\max}$, so once inflow subsides the storage decays exponentially and $\hat{\tau}_k = S_k^{\max}/C_k$

is the genuine e-folding recession constant, consistent with the exponential falling limbs fit in the preliminary gauge analysis ($\hat{\tau} = 37$ and 45 h). The drainage timescale $\hat{\tau}_k$ represents the expected duration of standing water under gravitational drainage and serves as the physically grounded prior mean on inundation duration; the model is designed so that any municipality with a USGS gauge ID and access to standard national GIS layers can parameterize it without field data collection. Sub-basins are delineated from 3DEP flow paths at the scale where storage and conveyance properties are resolvable from national layers, on the order of ten to twenty sub-basins across the flood-exposed core; a terrain-partition sensitivity analysis archived with the project repository indicates this granularity sits inside the plateau where finer partitioning adds degrees of freedom without adding explained building-level variation. Building counts per sub-basin are strongly skewed, from a dominant main-corridor sub-basin holding tens of buildings to single-building tributary sub-basins; the model is deliberately lumped at this scale, with per-building variation carried by the community adjustment δ_i^{comm} and the error variance rather than by finer hydraulics. Inundation duration at building i in sub-basin k is modeled as:

$$T_i \sim \text{LogNormal}(\mu_i,\ \sigma^2_{\text{MB}}) \tag{2}$$

where the log-scale prior mean is:

$$\mu_i \ = \ \log \hat{\tau}_k \ + \ \delta_i^{\text{comm}} \tag{3}$$

$\hat{\tau}_k$ is the mass balance drainage timescale for sub-basin k and σ^2_{MB} is the building-scale error variance of the GIS-parameterized mass balance. It absorbs both residual local drainage variability and the model's systematic limitations at building scale, so that all model error lives on the prior side of the framework rather than being attributed to the sensing data. σ^2_{MB} is a cross-sectional quantity, the building-to-building spread of durations within a single event, and it is estimated from the 2023 SAR brackets in Subtask 1.3, with the gauge record supplying only a weak regularizing anchor. A recession analysis of fifteen well-fit flood events in the gauge's 1990 to 2026 instantaneous record (archived with the project repository) gives an event-to-event variability of the watershed timescale, $\sigma_{\text{event}} = \text{sd}(\ln \hat{\tau}) = 0.30$. This is a storm-to-storm variability of one aggregate parameter, physically distinct from the within-event, building-to-building heterogeneity that σ_{MB} measures, so σ_{event} is not used as σ_{MB} itself but only as an order-of-magnitude anchor keeping the Subtask 1.3 calibration in a plausible range; an archived sensitivity study confirms the calibration is robust to mis-scaling of this anchor, with a factor-of-two mis-scaling moving the calibrated σ_{MB} by less than 0.01 under realistic acquisition cadences, and the 271 brackets carrying over 80 percent of the identifying weight even with a single usable post-peak scene. Subtask 1.2 delivers the GIS-only prior, setting $\delta_i^{\text{comm}} = 0$ gives the baseline model parameterized entirely from national GIS layers, which constitutes both the operational system for transfer communities and the comparison baseline for RQ2. The building-level adjustment δ_i^{comm}, elicited from community knowledge in Subtask 2.1 (locally known drainage obstructions, historically slow routes, anomalous building conditions), augments this baseline in Subtask 1.3; community knowledge improves the system but is not required to run it. The LogNormal family is appropriate because duration is strictly positive and right-skewed and keeps the log-scale consistency used by the Subtask 1.3 bracket likelihood. All duration quantities are conditional on inundation: T_i is the standing-water duration at building i given that it floods, and the Subtask 1.3 error model characterizes duration error, not flood/no-flood classification. Whether a building floods, and to what peak depth, is supplied by a second, deliberately simple computational component, a stage-to-DEM mapping that projects gauge crest stage onto the USGS 3DEP terrain model along a reach-scale water-surface slope:

$$d_i \ = \ \max\big\{\, 0,\ (s_g + \beta\, x_i) - z_i \big\} \tag{4}$$

where s_g is the crest water-surface elevation at the gauge, β is the reach-scale water-surface slope, x_i is the along-channel distance of building i upstream of the gauge, and z_i is the building's ground elevation from 3DEP; a building is predicted inundated where $d_i > 0$, so the same mapping supplies both building-level peak depth and the inundation extent. In post-event and planning use the projection takes observed or scenario crest stage, for example a recurrence of the 2023 crest; in pre-storm mode it takes the NWS river-stage forecast for the gauge. The 2023 ICEYE depth measurements calibrate the mapping's water-surface slope β and bound its building-level error in the same way the 2023 duration brackets calibrate the mass balance error, and that calibrated error defines a distribution over d_i rather than a single point value; the depth input required by the fragility surface of Subtask 1.4 in every operational mode is produced by this component, not by the mass balance model. A single reach-scale β is appropriate here even though the duration model resolves ten to twenty sub-basins. The flood water surface along the downtown reach is a smooth, hydraulically connected profile, a property of the reach as a whole rather than of any individual sub-basin, and a slope sensitivity analysis archived with the project repository bounds the admissible range of β from documented 2023 flood outcomes ahead of the ICEYE-based calibration. For pre-storm prediction, NOAA forecast precipitation replaces observed discharge as the model input; forecast uncertainty is propagated through the mass balance to produce appropriately wider credible intervals on predicted duration.

Subtask 1.3: Bayesian Fusion Framework (leads: Napolitano, Wauthier, and Busse). Subtask 1.3 formulates building-level inundation duration as a state-estimation problem, recovering a latent physical quantity from two information sources with two distinct error structures: the mass balance model, whose building-scale prediction error is carried by the prior variance σ^2_{MB}, and the SAR observations, whose uncertainty is a sampling property of the acquisition schedule rather than an instrument noise floor. A building's inundation duration is never observed directly. Each SAR overpass records only whether the building is inundated at that instant, so the observations constrain the true duration to a bracket,

$$T_i \;\in\; [L_i,\; U_i] \tag{5}$$

where L_i is the elapsed time to the last overpass at which building i is observed inundated and U_i the elapsed time to the first overpass at which it is observed dry; the bracket width is the revisit spacing of the acquired scenes. A building still observed wet at the final acquired scene is right-censored, with U_i unbounded; the same survival machinery applies, with likelihood $1 - \Phi((\ln L_i - \mu_i)/\sigma_{\mathrm{MB}})$. The likelihood of the observed wet/dry sequence is the prior probability mass on the bracket, $\Phi\left(\frac{\ln U_i - \mu_i}{\sigma_{\mathrm{MB}}}\right) - \Phi\left(\frac{\ln L_i - \mu_i}{\sigma_{\mathrm{MB}}}\right)$, and the posterior is the prior truncated to it:

$$T_i \mid [L_i, U_i] \;\sim\; \mathrm{LogNormal}(\mu_i,\; \sigma^2_{\mathrm{MB}}) \text{ truncated to } [L_i,\; U_i] \tag{6}$$

This interval-censored formulation treats the SAR observations honestly, with a dense acquisition sequence the brackets are hours wide and the posterior is sharply localized; with sparse acquisitions the brackets widen and the SAR data constrain the posterior only weakly. Per-scene wet/dry classification error ϵ_j, characterized by PI Wauthier in Subtask 1.1, generalizes the hard bracket to a product of per-overpass observation probabilities,

$$p(\boldsymbol{y}_i \mid T_i) \;=\; \prod_j (1-\epsilon_j)^{\,\mathbf{1}\{y_{ij}=\mathbf{1}\{t_j<T_i\}\}}\;\epsilon_j^{\,\mathbf{1}\{y_{ij}\neq\mathbf{1}\{t_j<T_i\}\}} \tag{7}$$

where y_{ij} is the wet/dry classification of building i at overpass time t_j; as $\epsilon_j \to 0$ this recovers the bracket of Equation (5), so classification error softens the truncation boundaries without changing the structure. σ^2_{MB} is calibrated on the 2023 event by maximum likelihood over the 271 interval-censored observations with the GIS-only prior ($\delta_i^{\mathrm{comm}} = 0$), using the classification-error-softened likelihood of Equation (7) with the per-scene error rates ϵ_j characterized in Subtask 1.1; this reduces to the hard-bracket survival likelihood where those error rates are small. Working on the log scale maintains dimensional consistency with the LogNormal prior and keeps all duration quantities strictly positive. The σ_{event} anchor centers a weak regularizing prior on σ_{MB} in this calibration, where the acquired brackets are wide, the censored likelihood constrains σ^2_{MB} only weakly and the estimate falls back toward the anchor rather than becoming unidentified, and the detectability statements reported in Subtask 2.3 make explicit what the acquired cadence can and cannot resolve. A pre-submission simulation study (archived with the project repository) quantifies this graceful degradation, across acquisition cadences from uniform six-hourly coverage down to a single usable post-peak scene, σ_{MB} remains identified with estimation uncertainty that grows continuously rather than collapsing, and the regularizing prior narrows the estimate at every cadence at a bias cost bounded by the anchor's own scale. Within this framework, the fusion that defines the computational core is between the physically grounded mass balance prior and the community knowledge adjustments δ_i^{comm} of Equation (3); the SAR brackets calibrate the error model on 2023 and adjudicate between prior variants on 2024, but the operational system never depends on them. Posteriors are computed independently per building; this is a simplification of the within-sub-basin spatial correlation of inundation durations, and its practical consequence is that portfolio-level uncertainty across neighboring buildings in the same sub-basin may be marginally underestimated; for the retrofit ranking decisions this proposal targets, priority rankings are robust to this approximation. For operational deployment without SAR imagery, the predictive distribution is the prior with the adjustment uncertainty folded in, $T_i \sim \mathrm{LogNormal}(\mu_i,\ \sigma_{\mathrm{MB}}^2 + P_i)$, where P_i is the tracked variance of the community adjustment from Subtask 2.4 and is zero in the GIS-only baseline; this variance carries the full building-scale error of the GIS-parameterized system together with the uncertainty of any elicited adjustment, the honest predictive uncertainty of the system when operating on free data alone. The 2024 Montpelier event is the held-out test. The calibrated framework, with all parameters fixed from 2023, is applied to 2024 gauge and NOAA forecast data and scored against the independently acquired 2024 SAR duration brackets.

Subtask 1.4: Duration-Dependent Fragility Surface Development (lead: Napolitano). Subtask 1.4 uses the processed ICEYE duration and depth for all 271 buildings (Subtask 1.1), runs in parallel with Subtasks 1.2 and 1.3, and does not depend on the mass balance model. Duration and peak depth are correlated in the Montpelier 2023 data, motivating a two-component approach that separates the duration effect physically before fitting it statistically. The *physical component* grounds the duration claim in the material science of how immersion time degrades older masonry, chiefly mortar softening, brick saturation, and rising damp, which sets the functional form of the duration effect independently of observational confounding. The *empirical component* calibrates the fragility surface parameters from field damage observations from the 2023 Montpelier flood, using ICEYE-measured depth and interval-censored ICEYE duration brackets as covariates at building scale for all 271 buildings; consistent with Subtask 1.3, duration enters the fitting not as a false point value but as its bracket, with estimation integrating over each building's bracket by Monte Carlo sampling from the truncated posterior of Equation (6). The fragility surface for damage state ds takes the form:

$$P(DS \geq ds \mid T_i,\ d_i) \ = \ \Phi\!\left(\frac{\ln T_i - \lambda_{0,ds} - \gamma \ln d_i}{\zeta}\right) \tag{8}$$

where T_i is inundation duration (hours), d_i is peak inundation depth (m), $\lambda_{0,ds}$ is the log-scale median duration capacity for damage state ds at reference depth, γ is the shared depth sensitivity (depth shifts the threshold at which duration governs damage), ζ is the shared log-standard deviation of capacity, and Φ is the standard normal cumulative distribution function. This is an ordered-probit (cumulative-probit) surface. The thresholds are constrained to be ordered, $\lambda_{0,1} < \lambda_{0,2} < \cdots$, while γ and ζ are shared across damage states, so the exceedance curves are horizontal translates of one another along the $\ln T_i$ axis and cannot cross at any (T_i, d_i); their successive differences therefore define a valid ordinal damage-state distribution everywhere. Estimation is Bayesian. The physical component supplies informative priors, with $\lambda_{0,ds}$ and γ centered at the material-science-derived values and their prior variances set from the published curves' specimen-to-specimen scatter, updated by the interval-censored 2023 field-damage likelihood. The resulting posterior surface is applied unchanged to 2024 for held-out validation. Published saturation-strength and capillary-absorption curves for historic brick and lime mortar [17–19] set these priors, with the immersion time at which strength loss crosses each damage state anchoring the ordered $\lambda_{0,ds}$, its acceleration with depth anchoring the shared γ, and specimen-to-specimen scatter anchoring the shared ζ. The claim under test is that duration adds explanatory power beyond depth alone. Because duration and depth are collinear in the 271-building sample, this claim is posed not as a naive coefficient t-test on the correlated data but as a comparison of posterior predictive accuracy on held-out 2024 damage between the bivariate model and a depth-only baseline. The Bayesian machinery makes both the identification and the significance claim coherent. The informative priors on $\lambda_{0,ds}$ and γ carry the identification the collinear 2023 covariates cannot, so the posterior shrinks toward the physical anchor where the empirical signal is weak and refines the anchored values in a genuinely data-driven way where it is strong. The strength of the duration-depth collinearity is quantified and reported directly, using variance inflation factors computed on the 2023 covariate data, and the bivariate versus depth-only comparison tests the difference in per-building held-out predictive scores with cluster-robust inference at the drainage sub-basin level, reflecting that the 271 buildings are geographically clustered within one downtown core rather than independently sampled; because the number of sub-basins is modest and the archived terrain-partition analysis shows a single main-corridor sub-basin holding roughly a third of the buildings, cluster-robust inference is implemented by wild cluster bootstrap, which remains valid with few and unbalanced clusters [39]. The posterior duration distribution from Subtask 1.3 (Equation (6)) propagates through Equation (8) by Monte Carlo integration, which samples d_i jointly with T_i from the calibrated depth distribution, so both fragility covariates carry quantified uncertainty into the decision interface and the resulting building-specific damage state probability distribution reflects the full inferential uncertainty of the mass balance, the depth projection, and the likelihood. A pre-submission sensitivity study (archived with the project repository) quantifies the relative contribution of depth-projection error and duration-bracket width to this uncertainty. The split is cadence dependent, with duration dominating where brackets are wide (depth error carries a median share of roughly four percent under the on-file 2023 acquisition) and depth-projection error becoming the dominant remaining source where dense six-hour brackets pin duration tightly, which is why both covariates carry propagated uncertainty rather than depth being treated as exact.

Subtask 1.5: System Validation (leads: all three PIs). Subtask 1.5 validates the full RO1 system against the 2024 held-out event across all three operational modes, with all model parameters fixed from 2023 calibration. Technical validation scores predictive duration distributions from the Bayesian framework against the 2024 SAR duration brackets for all 271 buildings, reporting the interval-censored log predictive score, empirical consistency of 90% credible intervals with the observed brackets (a well-calibrated system should have approximately 90% of brackets consistent with the corresponding intervals), and root mean square error against bracket midpoints where the acquisition cadence makes brackets tight; calibration in the strict frequentist sense requires repeated experiments, and the 271-building 2024 dataset constitutes one draw from the event distribution rather than independent trials, so coverage assessment from a single event is limited; the design uses all 271 buildings as assessment units while accounting for their geographic clustering through cluster-robust uncertainty estimates at the sub-basin level, consistent with Subtask 1.4, and multi-event calibration assessment is a target for future work. The interval-censored log predictive score for building i is

$$\mathrm{ILS}_i \;=\; -\log\left[\Phi\left(\frac{\ln U_i - \tilde{\mu}_i}{\tilde{\sigma}_i}\right) - \Phi\left(\frac{\ln L_i - \tilde{\mu}_i}{\tilde{\sigma}_i}\right)\right] \tag{9}$$

where $\tilde{\mu}_i$ and $\tilde{\sigma}_i$ are the predictive log-duration parameters for building i with all model components fixed from 2023; it is a proper scoring rule under interval censoring, and the same score adjudicates the prior variants compared in Subtask 2.3. Damage state validation compares Equation (8) fragility surface predictions against 2024 field damage observations at building scale, using the posterior duration distributions as input; the claim under test is that the duration-depth bivariate model predicts 2024 damage states more accurately than a depth-only baseline fitted on the same 2023 data. Because 2024 was a smaller-magnitude event, the statistical power of this held-out comparison depends on the damage variation the event actually produced; the comparison is therefore reported alongside a cross-validated bivariate versus depth-only comparison on the 2023 calibration data, and the 2024 result is stated relative to the observed 2024 damage-state distribution rather than as an unconditional significance claim. Decision-adequacy validation assesses whether retrofit priority rankings from the planning mode align with independent rankings produced by community partners and domain experts based on direct knowledge of 2024 damage outcomes; alignment is assessed by rank correlation, and systematic divergence between model rankings and partner rankings is treated as a finding requiring explanation rather than suppression. Pre-storm mode validation applies the framework retrospectively to the 2024 event using NOAA quantitative precipitation forecast products at 24-, 48-, and 72-hour lead times before the flood peak; predicted building-level inundation risk maps are compared against SAR imagery products, and prediction accuracy is reported as a function of lead time to characterize how forecast uncertainty degrades pre-storm prediction quality and what actionable lead time the system can reliably provide.

Key outcomes: RO1 produces three primary outputs: a validated Bayesian framework with characterized decision-adequacy across all three modes (Subtask 1.5); the duration-dependent fragility surfaces that give this CPS its physical grounding, covering the older load-bearing masonry these communities are built from (Subtask 1.4); and a characterization of when the GIS-parameterized system is decision-adequate, including the forecast lead times over which pre-storm predictions remain actionable (Subtask 1.5).

Research Objective 2 (RO2, lead: Napolitano with all PIs): Co-develop the CPS decision interface with community partners in Montpelier; characterize the conditions under which locally held community knowledge, entered as Bayesian calibration priors, improves damage assessment accuracy and calibration relative to GIS-only model runs; and produce a three-mode decision interface and transferable protocol for independent community use.

RO2 addresses RQ2 by characterizing the conditions under which community knowledge, entered as structural calibration input rather than post-hoc validation, improves the CPS outputs developed in RO1.

Subtask 2.1: Community Co-Development and Knowledge Elicitation (lead: Napolitano). PI Napolitano leads annual co-development workshops with emergency managers, historic preservation officials, and building owners in Montpelier across three years. Community partners include Vermont State Historic Preservation Office representatives, the Montpelier planning commission, the city emergency manager, and building owners in the historic downtown core; letters of collaboration from all partners are included in the supplementary materials. Building owners participate as primary knowledge sources. Their observations of which structures held water longest, which basement entries backed up, and which drainage paths ran clear constitute the building-level knowledge GIS cannot capture and that drives the δ_i^{comm} adjustments in Equation (3); community-engaged flood modeling likewise finds that tools succeed when anchored in local experiential knowledge [40, 41].

Year 1 workshops begin with a demonstration phase before elicitation. The team builds a working prototype of the planning-mode interface from the Subtask 1.2 GIS-only output and the 2023 ICEYE measurements, and presents it to partners to show what the GIS-only model knows and, critically, where it visibly errs relative to their direct experience; this demonstration-first structure earns the trust required for meaningful elicitation, and Subtask 2.2 interface co-development begins from this baseline in the same Year 1 workshops. Community-engaged flood technology research documents that residents of repeatedly flooded communities approach new tools with skepticism shaped by prior tools that promised help but required ongoing effort, and that these perceptions must be surfaced before engagement deepens; the demonstration-first structure addresses that skepticism before any knowledge commitment is asked [40, 42].

Following the demonstration phase, Year 1 surfaces locally held knowledge about which buildings held water longest, which drainage routes are historically slow, and which buildings have anomalous conditions; this knowledge is operationalized as δ_i^{comm} in Equation (3) through a structured two-step protocol. In the first step, the research team presents the GIS-only model predictions alongside the 2023 ICEYE duration map; community partners identify buildings where the model is visibly wrong and characterize the direction and approximate magnitude of the error (for example, "this drain always backs up, so this building holds water roughly twice as long as the model predicts"). In the second step, relative duration statements are converted to log-scale offsets: a building estimated to flood twice as long as the model predicts yields $\delta_i^{\mathrm{comm}} = \log 2 \approx 0.69$; half as long yields $\delta_i^{\mathrm{comm}} = -\log 2$; buildings where partners have no specific knowledge receive $\delta_i^{\mathrm{comm}} = 0$, falling back to the GIS-only prior. Each adjustment is recorded together with its stated physical mechanism (an obstructed culvert, a below-grade entry, a historically slow drainage route), not only its magnitude; mechanism attribution is what allows Subtask 2.3 to distinguish durable structural knowledge from recall of the error pattern the research team displayed. The uncertainty in each adjustment's magnitude is not absorbed into the prior's σ^2_{MB} term but is tracked explicitly as the variance P_i of the community adjustment, initialized from this elicitation and updated event by event through the closed-loop recursion of Subtask 2.4 (Equation (12)). The Year 2 cross-reference of community-stated adjustments against 2023 SAR actuals provides an empirical validity check on elicitation accuracy before the 2024 held-out test. Community partners also co-develop the criteria and weighting structure of the Subtask 2.2 decision interface and its two output tiers, shaping which investment outcomes matter most and what level of detail each audience needs; community-identified priorities are documented in structured form at Year 1 close to enable assessment of incorporation in subsequent years. *Year 2* cross-references community-observed damage against 2023 field damage states as an elicitation validity check, validates outputs against Year 1 stated priorities, and adds structured interpretation training so emergency managers and planning staff can read probabilistic outputs, run what-if simulations, interpret pre-storm risk maps, and present plain-language summaries to officials and residents without researcher support. *Year 3* stress-tests the interface for independent use across both tiers, reassesses whether community priorities have shifted, and evaluates whether managers can explain outputs without researcher involvement. This structured process constitutes the **CIR** component required by NSF 25-543: locally held knowledge enters the CPS as structural calibration input, community partners co-develop the framework architecture, and the process is assessed against documented community-identified priorities throughout.

Subtask 2.2: Three-Mode Decision Interface (lead: Napolitano). The decision interface co-developed with community partners supports three operational modes without requiring any knowledge of mass balance hydrology or Bayesian inference. Each mode produces two output tiers designed for distinct audiences, a design principle supported by both NSF SCC-funded community decision support research [40, 42] and federal guidance: NIST SP 1310 explicitly identifies the need for guidance that translates complex infrastructure assessments into simpler terms for decisionmakers and stakeholders who lack the technical background to interpret them directly [11, Sec. 6.1.7.4]. The *technical tier* presents probabilistic damage state distributions, credible intervals, ranked retrofit lists, and what-if outputs for emergency managers and planning staff; the *plain-language tier* translates the same analysis into a one-page building risk summary and a district-level investment comparison, for example, "Buildings on Main Street between Bridge and State have a 67% probability of damage requiring more than $15,000 in repairs in a flood similar to 2023; the proposed upstream retention basin reduces that to 43% across the district." Community partners co-design both tiers in Year 1 workshops.

In *planning mode* (always on, no flood required) the technical tier presents the ranked retrofit list and what-if simulations and the plain-language tier the one-page district comparisons. In *pre-storm mode* (triggered by NOAA forecast) the mass balance model and the Subtask 1.2 stage-to-DEM projection run on the NWS forecast crest, and the tiers output a building-level inundation risk map with credible intervals and a color-coded street map with a prioritized staging list that emergency managers use to pre-position barriers and issue targeted warnings. In *post-event mode* (after a flood), community members submit structured damage and drainage observations through the interface; these observations update δ_i^{comm} for the next event, and confirmed drainage retrofits trigger a recalculation of the affected sub-basin timescale $\hat{\tau}_k$ in the mass balance model. The planning and pre-storm modes are prototyped in Year 1 on the GIS-only prior; the probabilistic loss calculations below become operational in Year 2 once the Subtask 1.4 fragility surfaces and Subtask 1.3 posteriors are available. For each retrofit action a under consideration, the framework computes expected loss as a sum over buildings and damage states, each damage-state probability obtained by integrating the fragility surface of Equation (8) over the calibrated duration posterior of Equation (6) in Montpelier, or the predictive distribution of Subtask 1.3 where no SAR calibration exists:

$$E[\text{loss} \mid a] \;=\; \sum_{i} \sum_{ds=1}^{D} \left[\, P(DS_i \geq ds \mid a) - P(DS_i \geq ds{+}1 \mid a) \,\right] c_{ds}\, L_i \tag{10}$$

where L_i is the replacement value at building i, c_{ds} is the damage-state-specific loss ratio from HAZUS flood damage functions and FEMA benefit-cost damage ratios [9, 29], and the bracketed difference of successive exceedance probabilities is the probability that building i ends in exactly damage state ds, with the highest state D carrying no survivor term. Each $P(DS_i \geq ds \mid a)$ is computed per building by Monte Carlo integration of Equation (8) over building i's duration posterior jointly with its Subtask 1.2 depth distribution under the stage scenario evaluated, draws with no inundation contributing no loss; the outer sum runs over buildings with nonzero inundation probability under that scenario. The action a modifies both the fragility (building hardening raises the capacity threshold $\lambda_{0,ds}$) and the duration posterior (drainage investment reduces $\hat{\tau}_k$ through the mass balance model). Expected loss reduction is one criterion in the multi-criteria decision analysis score:

$$S(a) \;=\; \sum_{j=1}^{J} w_j \cdot V_j(a) \tag{11}$$

where criteria V_j include structural risk reduction (from Equation (10)), retrofit cost, historic preservation value, and community equity; weights w_j are co-developed with community partners in Year 1 workshops through structured pairwise comparison. Each V_j is min-max normalized to a common $[0, 1]$ scale over the candidate action set before weighting, so that w_j encodes the community's trade-off preference rather than an artifact of the criterion's raw units; the weighted sum is a compensatory method (a strong score on one criterion can offset a weak one on another), which is the intended behavior for this screening-level ranking of retrofit actions.

Subtask 2.3: Comparative Assessment of Community Knowledge Contribution (leads: all three PIs). Subtask 2.3 directly addresses RQ2 by running the full CPS under two conditions: with community-informed priors (δ_i^{comm} from Subtask 2.1 workshops) and with GIS-only priors ($\delta_i^{\text{comm}} = 0$), across all 271 buildings for both flood events. Predictive duration distributions under each condition are scored against the SAR brackets on the three Subtask 1.5 metrics (interval-censored log predictive score, 90% credible-interval consistency, and 2024 damage state accuracy). Because Year 1 elicitation deliberately shows partners the 2023 SAR map (Subtask 2.1), the 2023 comparison cannot distinguish independently held knowledge from accurate recall of the displayed error pattern; it is therefore treated as a diagnostic of elicitation mechanics rather than as a test of RQ2. The held-out 2024 event carries the substantive test. Adjustments that merely recall the 2023 error map improve 2024 predictions only if the underlying drainage anomalies persist across events, so cross-event persistence, evaluated separately for mechanism-attributed and unexplained adjustments, is the criterion that separates structural community knowledge from recall of what the research team displayed. Results are disaggregated by building type, drainage sub-basin, and 2023 damage state to characterize where community knowledge and geospatial data are substitutes versus complements. A building is community-knowledge-dominant if its δ_i^{comm} adjustment moves the 2024 predictive distribution meaningfully toward the observed brackets, GIS-dominant if it adds no value. The SAR bracket width bounds the signal this test can resolve, and the reported characterization states, per sub-group, what magnitude of δ_i^{comm} effect the available cadence can detect. A pre-submission simulation power study (archived with the project repository) supports the design targets. Under central assumptions the aggregate community-versus-GIS comparison reaches roughly 70 to 80 percent power at mechanism-attributed elicitation rates of 20 to 30 percent of the 271 buildings, power degrades with sparse acquisition cadence exactly as the bracket-width bound predicts, and elicited volume without mechanism quality contributes no power at any rate, confirming that the mechanism-attribution requirement in Subtask 2.1, not raw elicitation coverage, is what makes RQ2 testable. Cases where community knowledge conflicts with mass balance predictions and proves correct are the most informative outcomes. They identify where GIS cannot see the infrastructure conditions driving local drainage and constitute the primary evidence for elicitation value in future transfer. Cases where community knowledge underperforms the GIS-only baseline are reported in full, since the conditions under which elicitation adds no value are themselves a finding that informs the Subtask 2.4 transfer protocol.

Subtask 2.4: Closed-Loop Validation and Transferable Protocol (leads: all three PIs). The closed loop closes through Bayesian prior updating across events rather than post-retrofit physical detection. After each flood event, structured community observations submitted through the post-event mode update δ_i^{comm} for affected buildings; buildings whose observed durations consistently exceed or fall below mass balance predictions in the same direction receive updated prior means that carry forward to the next event. Formally, the per-event loop closes through a random-walk Kalman recursion on the community adjustment. Between events δ_i^{comm} follows a random walk with step variance q^2, $\delta_i^{\text{comm}} \to \delta_i^{\text{comm}} + \eta$, $\eta \sim \text{Normal}(0,\, q^2)$, because local drainage conditions are non-stationary; each event then updates both the estimate and its tracked variance P_i,

$$\begin{aligned} \delta_i^{\text{comm},(e+1)} &= \left(1 - w_i^{(e)}\right) \delta_i^{\text{comm},(e)} + w_i^{(e)} \left(\widehat{\ln T_i^{(e)}} - \ln \hat{\tau}_k\right), \qquad w_i^{(e)} = \frac{P_i^{(e)}}{P_i^{(e)} + v_i^{(e)}}, \\ P_i^{(e+1)} &= \left(1 - w_i^{(e)}\right) P_i^{(e)} + q^2 \end{aligned} \tag{12}$$

where $\widehat{\ln T_i^{(e)}}$ and $v_i^{(e)}$ are the posterior mean and variance of log duration for building i from event e, $w_i^{(e)}$ is the Kalman gain, and $P_i^{(e)}$ is the tracked variance of the current adjustment, initialized at $P_i^{(0)}$ from the Subtask 2.1 elicitation (buildings with no elicited knowledge start at the prior spread). The tracked variance converges within a few events to a steady state set by q, so where drainage conditions are stable the estimate becomes increasingly certain, while wide SAR brackets or vague post-event observations still produce small gains automatically; in the steady state the recursion behaves as a recency-weighted update whose decay rate on old information is set by q, appropriate for non-stationary drainage. An archived pre-submission simulation of this recursion confirms the convergence behavior and shows that a single cross-event pair constrains q only coarsely, so q is treated as a design hyperparameter with results reported across a swept range, coarsely bounded by the 2023-to-2024 persistence of the adjustments. After drainage retrofit completion, Co-PI Busse updates $\hat{\tau}_k$ directly from the engineering specifications of the completed work. A retention basin of known storage volume and outlet capacity changes the drainage timescale by a calculable amount, so the parameter is updated from design drawings rather than a full GIS model re-run; after building hardening retrofits, the fragility capacity threshold $\lambda_{0,ds}$ is updated for the retrofitted structures. No drainage retrofits are currently planned in Montpelier; partners are open to investment based on this research, and the Year 3 interface is designed to produce the drainage-versus-hardening comparison that would inform it. The retrofit update pathway is therefore demonstrated computationally within the grant. The team simulates a plausible drainage investment, updates $\hat{\tau}_k$ from its engineering specifications, and shows the resulting change in posterior durations and retrofit rankings. The 2024 event provides the first empirical test of whether one event cycle of community observation updates measurably improves posterior accuracy relative to the 2023-only calibration baseline. The transferable protocol specifies the minimum pathway for a new community: a USGS gauge ID, standard national GIS layer downloads (USGS 3DEP, NLCD, SSURGO, NHD StreamStats), and one co-development workshop to elicit initial δ_i^{comm} values, optionally supported by free Sentinel-1 imagery as a rough visual aid for flood-extent discussion where local recall is difficult, combined with the open-source software package initialized from

Montpelier-derived mass balance and fragility parameters. A community operating under this protocol has wider credible intervals than Montpelier because it lacks local ICEYE calibration, and free Sentinel-1 imagery does not close this gap. The expected interval widths, derived from the Subtask 1.3 σ^2_{MB} characterization and the elicitation-initialized $P_i^{(0)}$, are reported in the protocol documentation so communities can assess decision-adequacy before adoption, and intervals narrow with each observation cycle.

Key outcomes: RO2 produces three primary outputs: a three-mode interface independently operated by Montpelier partners in Year 3 (Subtask 2.2); a characterization of when community knowledge and geospatial data are substitutes versus complements, including cases where community knowledge adds no value (Subtask 2.3); and a transferable protocol enabling any municipality with a USGS gauge to initialize the system from one workshop without a commercial SAR purchase (Subtask 2.4).

5 Broader impacts: The closed-loop CPS framework gives small, resource-constrained municipalities probabilistic damage assessments and pre-storm decision support for retrofit and drainage investment decisions, a pathway that does not currently exist. By enabling better-informed retrofit decisions, the framework reduces the flood damage burden that falls disproportionately on towns with the fewest resources to recover. The planning mode is available at deployment; pre-storm mode delivers actionable building-level risk before water rises; post-event mode improves accuracy with each event, all without proprietary data or ongoing research support.

The framework's operational data inputs are only USGS gauge data and NOAA precipitation, both free and nationally available; the mass balance model is parameterizable from USGS 3DEP, NLCD, SSURGO, and NHD StreamStats for any municipality with a gauge. A new community needs a gauge ID, standard GIS access, and one community workshop to elicit local drainage knowledge; no commercial SAR purchase, no hydrologist, and no hydraulic model is required.

Emergency managers, historic preservation officials, and planning staff gain structured capacity to interpret probabilistic damage assessments, use forecast inputs for pre-event staging, and run what-if investment simulations independently. The CIR co-development methodology, documented as a transferable protocol, advances CPS practice by showing how structured elicitation of locally held flood knowledge translates into informative Bayesian priors, making resident knowledge measurably load-bearing in model calibration rather than supplementary narrative. Graduate students will receive cross-disciplinary training spanning SAR imagery data processing (Co-PI Wauthier), hydrological mass balance modeling and GIS-parameterized watershed analysis (Co-PI Busse), and probabilistic structural assessment and community-engaged research (PI Napolitano).

5.1 Potential for transformative impact: This framework is the first to place community-scale drainage investment and individual building retrofits on the same probabilistic basis for small historic municipalities, resolving an allocation problem that annually re-injures them. Its generalizable principle, that free national data fused with community knowledge suffices for the investment decisions these municipalities face, extends beyond flood resilience to any infrastructure domain with national data layers and community knowledge but no simulation capacity, and its substitutes-versus-complements characterization (Key Outcomes, RO2) informs how CPS research treats locally held knowledge more broadly.

5.2 Education and broadening participation: In addition to broader impacts for study communities, the project integrates research outcomes into engineering education.

Undergraduate students hired for this project will participate in the Building Engineering Seminal Undergraduate Research Experience (BESURE) program at Penn State

5.3 Open science and transferability: The Bayesian fusion code, mass balance parameterization pipeline, and decision interface will be released on GitHub under an MIT license, with documentation enabling any municipality to initialize the system from national GIS data layers. Derived SAR imagery, codes, and building-level inundation products for Montpelier will be shared on DesignSafe-CI; field damage state observations and community workshop documentation will be shared on the Open Science Framework with appropriate anonymization.

6 Project timeline and evaluation: Progress is monitored through monthly PI meetings and annual reviews. RO1 technical development (Subtasks 1.1 to 1.4) runs in Years 1 and 2 parallel to RO2 Year 1 co-development and interface construction; validation (Subtask 1.5) and comparative assessment (Subtask 2.3) follow in Year 2; closed-loop validation, protocol transfer (Subtask 2.4), and Year 3 independent-use stress testing conclude in Year 3. Success is evaluated across three dimensions via the following key performance indicators:

- *Technical KPIs* (Subtask 1.5):
  - Building-level duration error against 2024 SAR brackets at or below the ranking-stability threshold (pre-submission sensitivity analysis, archived with the repository): $\sim$0.4 log-units / $\sim$70 h RMSE, below which the top-20 retrofit ranking retains $\geq$80% overlap with the error-free ranking.
  - Empirical consistency of 90% credible intervals with the observed brackets between 0.85 and 0.95 across 271 buildings.
  - Duration-depth bivariate model accuracy exceeding the depth-only baseline by a margin statistically significant under cluster-bootstrap comparison of per-building predictive scores, in 2023 cross-validation and on the 2024 held-out event relative to that event's smaller damage variation.
  - Pre-storm mode producing actionable building-level risk rankings at 48-hour forecast lead time.
- *Community engagement KPIs* (Subtask 2.1):
  - Proportion of Year 1 community priorities reflected in Year 2 framework outputs, evaluated by community partners in Year 2 workshops, with discrepancies documented and addressed before Year 3.
  - Structured elicitation covering all 271 buildings by end of Year 1, yielding nonzero, mechanism-attributed δ_i^{comm} adjustments for at least 20% of buildings (target 30%, the power threshold established in Subtask 2.3); zero defaults are recorded as explicit no-knowledge outcomes, not elicited values.
- *Community knowledge contribution KPIs* (Subtask 2.3): community-informed model achieves a better interval-censored log predictive score and better calibration than the GIS-only baseline on at least one disaggregated sub-group; substitutes-versus-complements characterization published with full results including null findings.
- *Adoption KPIs* (Year 3): emergency managers operate all three decision interface modes without researcher support in the Year 3 stress test, and demonstrate they can explain pre-storm risk maps and planning-mode retrofit comparisons to elected officials and residents unaided.

7 Results from prior NSF Support: PI Napolitano: PI Napolitano was PI on NSF CMMI-2222849 ($131,317, 04/2022–02/2025): "RAPID: Examining the Performance of Historic Masonry Structures after the Midwest Tornadoes." Intellectual Merit: This project addressed building code gaps by analyzing historic masonry structures' performance under tornadic loading and assessing retrofitting efficacy, producing the first systematic dataset on tornado damage to unreinforced masonry heritage structures. Broader Impacts: The project enhanced understanding of historic masonry resilience under wind hazards, produced open datasets archived on DesignSafe, and integrated community-centered structural data into engineering education. Publications: [43–48]. Undergraduate research presentations: [49, 50]. Publicly shared datasets: [51–54]. Renewal: N/A. **Co-PI Wauthier:** Christelle Wauthier is the PI on NSF EAR-1945417 ($499,999k, 08/01/20-07/31/26): "CAREER: Numerical Modeling of Volcanic Flank Instability Processes". Intellectual Merit: This CAREER proposal addresses why the style, behavior, and timing of the cyclic growth then destruction of volcanic edifices varies across such a wide spectrum. It focuses on flank collapse on ocean island volcanoes and in using modeling to unravel the observed complexity in data sets representing a spectrum of type volcanoes. Broader Impacts: Graduate and undergraduate

education, support for underrepresented undergraduate research interns, new educational activities, and FEM and other modeling training for graduate students. Research products: Cooper et al., 2023; Leeburn* et al., 2022; Gonzalez-Santana* et al., 2022, 2024; Gonzalez-Santana* and Wauthier, 2021; Poppe* et al., 2024; Kim* et al., 2024; Wauthier and Ho, 2024; Gonzalez-Santana* et al., 2025, all of which are listed in the References.
Co-PI Busse: Margaret Busse is senior personnel on NSF CBET-2400672 ($1,000,000, 8/2024 - 7/2029): "RAISE: CET: Co-development of engineering energy literacy and engineering identities through campus decarbonization research." Intellectual Merit: This work integrates six component-specific research directions to address fundamental questions associated with net-zero fuels in the context of Combined Heat and Power (CHP) decarbonization. Intellectual Merit of the research lies in the integrated real-world system evaluation approach as well as the evaluation of student energy literacy through their research experience. Broader Impacts: The Broader Impacts include advancement of decarbonization technologies and the development of an energy-literate future engineering workforce. Research Products: None.

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